STA2002 Tutorial 8

1 Review

Chi-square goodness of fit test

Suppose we have data $X_1, \ldots, X_n$ that may fall in k categories, say $A_1, A_2, \ldots, A_k$, and Y_i is the count of data that is in category i for $i = 1, 2, \ldots, k$. Let $p_i = P(X \in A_i)$, and we would like to test

$$H_0 : p_i = p_{i0}, \quad i = 1, 2, \ldots, k \quad H_1 : H_0 \text{ is not true}$$

Under H_0, the test statistic:

$$Q_{k-1} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = \sum_{i=1}^{k} \frac{(Y_i - np_{i0})^2}{np_{i0}} \sim \chi^2(k-1)$$

we reject H_0 at level α when

$$q_{k-1} \geq \chi^2_\alpha(k-1)$$

Degree of freedom $k-1$: We have $k-1$ free parameters p_i ($p_k = 1 - p_1 - p_2 - \ldots p_{k-1}$) to describe the model.

If the test contains **unspecified parameter**, i.e.

$$H_0 : (Y_1, \ldots, Y_k) \sim \text{Multinomial}(n, p_1(\theta), \ldots, p_k(\theta)) \quad H_1 : H_0 \text{ is not true}$$

where $\theta \in \mathbb{R}^d$ ($d < k$). We first need to estimate θ by maximizing the likelihood

$$L(\theta | y_1, \ldots, y_k) = (p_1(\theta))^{y_1} \ldots (p_k(\theta))^{y_k}$$

and then

$$Q = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = \sum_{i=1}^{k} \frac{(Y_i - np_i(\hat{\theta}))^2}{np_i(\hat{\theta})} \sim \chi^2(k-1-d)$$

Degree of freedom $k-1-d$: we have $k-1$ free parameters to describe the model and under H_0 there are d parameters (in $\hat{\theta}$) to be estimated.

Chi-square test for homogeneity

Suppose we have h groups and k categories. Observe data $x_1, \ldots, x_{n_j}$ from group $j \in \{1, 2, \ldots, h\}$. The data may fall in category i, $i \in \{1, 2, \ldots, k\}$. Let

$$p_{ij} = P(\text{Observe data from category } i | \text{ data is from group } j)$$

and Y_{ij} be the corresponding counts. $\sum_i p_{ij} = 1$

	1	2	...	h	
1	y_{11}				
2					
$\vdots$			y_{ij}		
k					
	n_1		...	n_h	n

$$H_0 : p_{ij} = p_i, \quad i = 1, 2, \ldots, k \quad H_1 : H_0 \text{ is not true}$$

Define

$$\hat{p_i} = \frac{\sum_{j=1}^{h} Y_{ij}}{\sum_{j=1}^{h} n_j}, \quad i = 1, 2, \ldots, k$$

Under H_0, the test statistic:

$$\sum_{j=1}^{h}\sum_{i=1}^{k} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \sum_{j=1}^{h}\sum_{i=1}^{k} \frac{(Y_{ij} - n_j\hat{p}_i)^2}{n_j\hat{p}_i} \sim \chi^2(h(k-1) - (k-1))$$

Degree of freedom: we have $k-1$ free parameters to describe the model for each group, h groups. Under H_0 there are $k-1$ free parameters to estimate.
we reject H_0 at level α when

$$q \geq \chi^2_\alpha((h-1)(k-1))$$

Chi-square test for Independence

We have h groups and k categories. Observe pooled data $x_1, \ldots, x_n$. Let

$$p_{ij} = P(\text{data is from group } j \text{ and from category } i)$$

and Y_{ij} the corresponding counts. $\sum_i \sum_j p_{ij} = 1$

	1	2	...	h	
1	y_{11}				$n_{1\bullet}$
2					
$\vdots$			y_{ij}		$\vdots$
k					$n_{k\bullet}$
	$n_{\bullet 1}$		...	$n_{\bullet h}$	n

$$H_0 : p_{ij} = p_{i\bullet}p_{\bullet j}, \quad i = 1, 2, \ldots, k \quad H_1 : H_0 \text{ is not true}$$

Define

$$\hat{p}_{\bullet j} = \frac{\sum_{i=1}^{k} Y_{ij}}{n} = \frac{y_{\bullet j}}{n}$$

$$\hat{p}_{i\bullet} = \frac{\sum_{j=1}^{h} Y_{ij}}{n} = \frac{y_{i\bullet}}{n}$$

Under H_0, the test statistic:

$$\sum_{j=1}^{h}\sum_{i=1}^{k} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = \sum_{j=1}^{h}\sum_{i=1}^{k} \frac{(Y_{ij} - n\hat{p}_{\bullet j}\hat{p}_{i\bullet})^2}{n\hat{p}_{\bullet j}\hat{p}_{i\bullet}} \sim \chi^2(hk - 1 - (k-1) - (h-1))$$

Degree of freedom: we have $hk-1$ free parameters to describe the model and under H_0 there are $(k-1)+(h-1)$ free parameters to estimate.
we reject H_0 at level α when

$$q \geq \chi^2_\alpha((h-1)(k-1))$$

1. (**Chi-square test**) It has been claimed that, for a penny minted in 1999 or earlier, the probability of observing heads upon spinning the penny is $p = 0.30$. Three students got together, and they would each spin a penny and record the number X of heads out of the three spins. They repeated this experiment $m = 200$ times, observing 0, 1, 2, and 3 heads 57, 95, 38, and 10 times, respectively.

   (a) Use these data to test the hypotheses that X is $b(3, 0.30)$.

   (b) Give a range for the p-value of this test.

   (c) In addition, out of the 600 spins, calculate the number of heads occurring and then a 95% confidence interval for p.

   (a) Using Table II in Appendix B with $p = 0.30$, the hypothesized probabilities are

$$p_0 = P(X = 0) = 0.343,$$

$$p_1 = P(X = 1) = 0.441$$

$$p_2 = P(X = 2) = 0.189$$

$$p_3 = P(X = 3) = 0.027$$

   Thus the respective expected values are 68.6, 88.2, 37.8, and 5.4. The value of the chi-square goodness of fit statistic is:

$$q = \frac{(57 - 68.6)^2}{68.6} + \frac{(95 - 88.2)^2}{88.2} + \frac{(38 - 37.8)^2}{37.8} + \frac{(10 - 5.4)^2}{5.4} = 6.405 < 7.815 = \chi^2_{0.05}(3)$$

   Do not reject the hypothesis that X is $b(3, 0.30)$ at a 5% significance level.

   (b) The range for the p-value is

$$0.05 < \text{p-value} < 0.10$$

   because $\chi^2_{0.10}(3) = 6.251 < 6.405 < \chi^2_{0.05}(3) = 7.815$

   (c) We find that $\hat{p} = 201/600 = 0.335$. Thus a 95% confidence interval for p is

$$0.335 \pm 1.96\sqrt{(0.335)(0.665)/600} \text{ or } [0.297, 0.373]$$

2. (**Test for homogeneity**) Suppose that three comparable classed of 15 students responded to three different methods of instructions, giving the following scores on a standardized test:
Classe U:
91 42 39 62 55 82 67 44 51 77 61 52 76 41 59
Class V:
80 71 55 67 61 93 49 78 57 88 79 81 63 51 75
Class W:
91 73 67 83 59 98 87 69 78 80 65 94 82 74 85
Use a chi-square test with $\alpha = 0.05$ to test the equality of the three distribution by dividing the combined sample into three categories. In the combined sample of 45 observations, the lower third includes those with scores of 61 or lower, the middle third have scores from 62 through 78 , and the higher third are those with scores of 79 and above.

   In the combined sample of 45 observations, the lower third includes those with scores of 61 or lower, the middle third have scores from 62 through 78, and the higher third are those with scores of 79 and above.

$$q = 3.2 + 0.2 + 1.8 + 0 + 0 + 0 + 3.2 + 0.2 + 1.8 = 10.4 > 9.488 = \chi^2_{0.05}(4)$$

   we reject the equality of these three distributions.

	low	middle	high	totals
class U	9	4	2	15
expected	5	5	5	
class V	5	5	5	15
expected	5	5	5	
class W	1	6	8	15
expected	5	5	5	
totals	15	15	15	45

3. (**Test for independence**) A random survey of 100 students asked each student to select the most preferred form of recreational activity from five choices. Following are the results of the survey:

	Recreational Choice					
Gender	Basketball	Baseball Softball	Swimming	Jogging Running	Tennis	Totals
Male	21	5	9	12	13	60
Female	9	3	1	15	12	40
Totals	30	8	10	27	25	100

Test whether the choice is independent of the gender of the respondent. Approximate the p-value of the test. Would we reject the null hypothesis at $\alpha = 0.05$?

	Recreational Choice					
Gender	Basketball	Baseball softball	Swimming softball	Jogging Running	Tennis	Totals
Male	21	5	9	12	13	60
Expected	(18)	(4.8)	(6)	(16.2)	(15)	
Female	9	3	1	15	12	40
Expected	(12)	(3.2)	(4)	(10.8)	(10)	
Totals	30	8	10	27	25	100

$$q = \frac{(21-18)^2}{18} + \frac{(5-4.8)^2}{4.8} + \frac{(9-6)^2}{6} + \cdots + \frac{(1-4)^2}{4} + \frac{(15-10.8)^2}{10.8} + \frac{(12-10)^2}{10} = 8.41 < 9.488 = \chi^2_{0.05}(4)$$

Do not reject H_0.